Deep Learning Foundations for GenAI & Agentic AI

Session Flow

- Problem-first approach
- Neural Networks
- Loss & Optimization
- Overfitting
- Attention
- Transformers → GenAI

Business Problem

- Why is customer churn increasing?
- What data needed?
- How to model?

Neural Network Basics

- Input → Weights → Activation → Output
- Layers and neurons
- Learn via weights

Hands-on NN

- Build simple NN
- Train on dummy dataset
- Observe outputs

Loss Functions

- MSE for regression
- Cross entropy for classification
- Goal: minimize error

Optimization

- Gradient descent
- Learning rate impact
- Iterative improvement

Overfitting

- Low train loss, high val loss
- Memorization vs learning
- Fix: dropout, more data

Attention Intuition

- Focus on important inputs
- Dynamic weighting
- Context understanding

Self Attention

- Q, K, V concept
- Compute similarity
- Weighted output

Transformers

- Self-attention + feedforward
- Parallel processing
- Context-aware

GenAI Bridge

- Transformers → LLMs
- LLMs → Agents
- Agents → decision systems

Wrap Up

- NN → Attention → Transformers
- Foundation for GenAI
- Next: Agentic AI